

Stat 517
Homework # 5

1. Let $X_1, X_2, \dots, X_n$ be a random sample from a $\text{Gamma}(\alpha = 3, \beta = \theta)$ distribution, $0 < \theta < \infty$. Determine the MLE of θ .
2. Let $Y_1 < Y_2 < \dots < Y_n$ be the order statistics of a random sample from a distribution with pdf $f(x; \theta) = 1, \theta - \frac{1}{2} \leq x \leq \theta + \frac{1}{2}, -\infty < \theta < \infty$, zero elsewhere. This is a nonregular case. Show that every statistics $u(X_1, \dots, X_n)$ such that

$$Y_n - \frac{1}{2} \leq u(X_1, \dots, X_n) \leq Y_1 + \frac{1}{2}$$

is a mle of θ . In particular, $(4Y_1 + 2Y_n + 1)/6$, $(Y_1 + Y_n)/2$, and $(2Y_1 + 4Y_n - 1)/6$ are three such statistics. Thus, uniqueness is not, in general, a property of mles.

3. Suppose $X_1, \dots, X_n$ are iid with pdf $f(x; \theta) = 2x/\theta^2, 0 < x \leq \theta$, zero elsewhere. Note this is a nonregular case. Find:
 - (a) The mle of $\hat{\theta}$ for θ .
 - (b) The constant c so that $E(c\hat{\theta}) = \theta$.
 - (c) The mle for the median of the distribution. Show that it is a consistent estimator.
4. Suppose $X_1, X_2, \dots, X_n$ are iid with pdf $f(x; \theta) = \theta^{-1}e^{-x/\theta}, 0 < x < \infty$, zero elsewhere. Find the mle of $P(X \leq 2)$ and show that it is consistent.

1. Let $X_1, X_2, \dots, X_n$ be a random sample from a $\text{Gamma}(\alpha = 3, \beta = \theta)$ distribution, $0 < \theta < \infty$. Determine the MLE of θ .

• scale-form Gamma density:

$$f(x; \alpha, \beta) = \frac{x^{\alpha-1} e^{-x/\beta}}{\Gamma(\alpha) \beta^\alpha}, \quad x > 0 \quad \rightarrow \quad f(x; \theta) = \frac{x_i^2 e^{-x_i/\theta}}{\Gamma(3) \theta^3} = \frac{x_i^2 e^{-x_i/\theta}}{2\theta^3}$$

• Likelihood Function:

$$\begin{aligned} L(\theta) &= \prod_{i=1}^n f(x_i; \theta) = f(x_1; \theta) \cdot f(x_2; \theta) \dots f(x_n; \theta) = \prod_{i=1}^n \left[\frac{x_i^2 e^{-x_i/\theta}}{2\theta^3} \right] \\ &= \left(\prod_{i=1}^n x_i^2 \right) \left(\prod_{i=1}^n \frac{1}{2} \right) \left(\prod_{i=1}^n \frac{1}{\theta^3} \right) \left(\prod_{i=1}^n e^{-x_i/\theta} \right) = \frac{1}{2^n} \cdot \frac{1}{\theta^{3n}} \cdot \frac{1}{\theta^6} \cdot e^{-(x_1 + x_2 + \dots + x_n)/\theta} \\ &= \left(\left(\prod_{i=1}^n x_i^2 \right) / 2^n \cdot \theta^{3n} \right) \exp\left(-\frac{1}{\theta} \sum_{i=1}^n x_i\right) \end{aligned}$$

$\underbrace{e^{-(x_1 + x_2 + \dots + x_n)/\theta}}_{\sum_{i=1}^n e^{-(x_i)/\theta}} = \exp\left(-\frac{1}{\theta} \sum_{i=1}^n x_i\right)$

• Log Likelihood:

$$\begin{aligned} \ell(\theta) &= \log L(\theta) = \log \left(\left(\prod_{i=1}^n x_i^2 \right) / 2^n \cdot \theta^{3n} \right) \exp\left(-\frac{1}{\theta} \sum_{i=1}^n x_i\right) \\ &= \log \left(\prod_{i=1}^n x_i^2 \right) - \log(2^n) - \log(\theta^{3n}) + \log \left[\exp\left(-\frac{1}{\theta} \sum_{i=1}^n x_i\right) \right] \\ &= \underbrace{2 \sum_{i=1}^n \log x_i}_{\text{constant}} - \underbrace{n \log 2}_{\text{constant}} - 3n \log \theta - \frac{1}{\theta} \sum_{i=1}^n x_i = -3n \log \theta - \frac{1}{\theta} \sum_{i=1}^n x_i + C \\ \ell'(\theta) &= \frac{d}{d\theta} [-3n \log \theta] - \frac{d}{d\theta} \left[\frac{1}{\theta} \sum_{i=1}^n x_i \right] + \cancel{C} = \frac{-3n}{\theta} + \frac{\sum_{i=1}^n x_i}{\theta^2} \end{aligned}$$

$\downarrow \quad \quad \quad \downarrow \quad \quad \quad \downarrow$
 $\frac{d}{d\theta} \log \theta = \frac{1}{\theta} \quad \quad -(\sum x_i) \theta^{-1} \quad \quad \frac{d}{d\theta} C = 0$
 $\quad \quad \quad \downarrow$
 $\frac{d}{d\theta} \theta^{-1} = \theta^{-2}$

• solve for $\hat{\theta}$:

$$\begin{aligned} \times (\theta^2) \quad \frac{-3n}{\theta} + \frac{\sum_{i=1}^n x_i}{\theta^2} &= 0 \times (\theta^2) \\ -3n\theta + \sum x_i &= 0 \rightarrow \frac{\sum x_i}{3n} = \frac{3n\theta}{3n} \rightarrow \hat{\theta} = \frac{\sum x_i}{3n} \end{aligned}$$

using $\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$,
 $\sum_{i=1}^n x_i = \bar{x} n$

$$\hat{\theta}_{MLE} = \frac{\bar{x}}{3}$$

- $Y_1 = \min(X_1, \dots, X_n)$ smallest obs.
- $Y_n = \max(X_1, \dots, X_n)$ largest obs.

2. Let $Y_1 < Y_2 < \dots < Y_n$ be the order statistics of a random sample from a distribution with pdf $f(x; \theta) = 1, \theta - \frac{1}{2} \leq x \leq \theta + \frac{1}{2}, -\infty < \theta < \infty$, zero elsewhere. This is a nonregular case. Show that every statistics $u(X_1, \dots, X_n)$ such that

$$X \sim \text{Uniform}(\theta - \frac{1}{2}, \theta + \frac{1}{2}) \quad Y_n - \frac{1}{2} \leq u(X_1, \dots, X_n) \leq Y_1 + \frac{1}{2}$$

is a mle of θ . In particular, $(4Y_1 + 2Y_n + 1)/6$, $(Y_1 + Y_n)/2$, and $(2Y_1 + 4Y_n - 1)/6$ are three such statistics. Thus, uniqueness is not, in general, a property of mles.

$(\theta - \frac{1}{2}, \theta + \frac{1}{2})$ Interval

- Midpoint = $\theta : \frac{(\theta - \frac{1}{2}) + (\theta + \frac{1}{2})}{2} = \theta$

- Interval width: $(\theta + \frac{1}{2}) - (\theta - \frac{1}{2}) = 1$

- $L(\theta) = \prod_{i=1}^n f(x_i; \theta) = \prod_{i=1}^n I\{\theta - \frac{1}{2} \leq x_i \leq \theta + \frac{1}{2}\}$

$I = 1$ if all obs. are inside the interval.

$$f(x; \theta) = I\{\theta - \frac{1}{2} \leq x \leq \theta + \frac{1}{2}\}$$

$I = 1$ if statement is true (inside the interval)

- Lower endpoint cannot be greater than the smallest Observation:

$$\theta - \frac{1}{2} \leq Y_1 \rightarrow \theta \leq Y_1 + \frac{1}{2}$$

- Upper endpoint must be at least / greater than the largest Observation:

$$\theta + \frac{1}{2} \geq Y_n \rightarrow \theta \geq Y_n - \frac{1}{2}$$

$$Y_n - \frac{1}{2} \leq \hat{\theta} \leq Y_1 + \frac{1}{2}$$

$\downarrow$
 $u(X_1, \dots, X_n)$

For every statistic $u(X_1, \dots, X_n)$, that satisfies, it is an MLE of θ .

$$Y_n - Y_1 \leq 1$$

Distance between largest + smallest obs. cannot be greater than 1.

$\downarrow$ continue

weighted Average: $w(Y_n - \frac{1}{2}) + (1-w)(Y_1 + \frac{1}{2})$, $0 \leq w \leq 1$

$$\begin{aligned} \textcircled{1} (4Y_1 + 2Y_n + 1)/6 \\ = \frac{4Y_1}{6} + \frac{2Y_n}{6} + \frac{1}{6} \\ = \frac{2}{3}Y_1 + \frac{1}{3}Y_n + \frac{1}{6} \\ \rightarrow \frac{1}{3}(Y_n - \frac{1}{2}) + \frac{2}{3}(Y_1 + \frac{1}{2}) \\ = \frac{1}{3}Y_n - \frac{1}{6} + \frac{2}{3}Y_1 + \frac{1}{3} \\ = \frac{1}{3}Y_n + \frac{2}{3}Y_1 + \frac{1}{6} = \frac{2Y_n + 4Y_1 + 1}{6} \end{aligned}$$

$$\begin{aligned} \textcircled{2} (Y_1 + Y_n)/2 \\ \rightarrow \frac{1}{2}(Y_n - \frac{1}{2}) + \frac{1}{2}(Y_1 + \frac{1}{2}) \\ = \frac{1}{2}Y_n - \frac{1}{4} + \frac{1}{2}Y_1 + \frac{1}{4} \\ = \frac{1}{2}Y_n + \frac{1}{2}Y_1 = \frac{Y_1 + Y_n}{2} \end{aligned}$$

$$\begin{aligned} \textcircled{3} (2Y_1 + 4Y_n - 1)/6 \\ = \frac{1}{3}Y_1 + \frac{2}{3}Y_n - \frac{1}{6} \\ \rightarrow \frac{2}{3}(Y_n - \frac{1}{2}) + \frac{1}{3}(Y_1 + \frac{1}{2}) \\ = \frac{2}{3}Y_n - \frac{1}{3} + \frac{1}{3}Y_1 + \frac{1}{6} \\ = \frac{2}{3}Y_n + \frac{1}{3}Y_1 - \frac{1}{6} = \frac{4Y_n + 2Y_1 - 1}{6} \end{aligned}$$

- Each expression is a weighted average of $(Y_n - \frac{1}{2})$ and $(Y_1 + \frac{1}{2})$, so each lies in $[A, B]$. Hence, all 3 statistics are MLEs of θ .
- MLE is not unique; not a property of MLEs.

3. Suppose $X_1, \dots, X_n$ are iid with pdf $f(x; \theta) = 2x/\theta^2$, $0 < x \leq \theta$, zero elsewhere. Note this is a nonregular case. Find:

(a) The mle of $\hat{\theta}$ for θ .

(b) The constant c so that $E(c\hat{\theta}) = \theta$.

(c) The mle for the median of the distribution. Show that it is a consistent estimator.

$$\downarrow \\ x_i \leq \theta, \text{ and } x_{(n)} \leq \theta \\ x_{(n)} = \max(X_1, \dots, X_n)$$

$$a) f(x_i; \theta) = \frac{2x_i}{\theta^2} I\{0 < x_i \leq \theta\}$$

$$L(\theta) = \prod_{i=1}^n \frac{2x_i}{\theta^2} I\{x_i \leq \theta\} = \frac{2^n \prod_{i=1}^n x_i}{\theta^{2n}} I\{x_{(n)} \leq \theta\} \rightarrow \begin{aligned} I &= 1 \text{ if } x_{(n)} \leq \theta \\ I &= 0 \text{ if } x_{(n)} > \theta \end{aligned}$$

$$\text{For values } \theta > x_{(n)}: L(\theta) = \frac{2^n \prod_{i=1}^n x_i}{\theta^{2n}} \text{ constant}$$

$$L(\theta) \propto \frac{1}{\theta^{2n}} \text{ As } \theta \uparrow, L(\theta) \downarrow.$$

Therefore, we choose the smallest value of θ :

$$\hat{\theta} = x_{(n)} = \max(X_1, \dots, X_n)$$

$\downarrow$ continue

$$b) E(c\hat{\theta})=0 \rightarrow cE(\hat{\theta})=0 \rightarrow cE(X_{(n)})$$

$$\hat{\theta} = X_{(n)}$$

$$\bullet \text{CDF: } F(x) = P(X \leq x) = \int_{-\infty}^x f(t|\theta) dt$$

$$F(x) = \int_0^x \frac{2t}{\theta^2} dt = \frac{1}{\theta^2} \int_0^x \underbrace{2t}_{t^2} dt = \frac{1}{\theta^2} \underbrace{[t^2]_0^x}_{x^2} = \frac{x^2}{\theta^2}, \quad 0 < x \leq \theta$$

$$\bullet P(X_{(n)} \leq x) = P(X_1 \leq x, \dots, X_n \leq x) = \prod_{i=1}^n P(X_i \leq x)$$

$$F_{X_{(n)}}(x) = [F(x)]^n = \left(\frac{x^2}{\theta^2}\right)^n = \left(\frac{x}{\theta}\right)^{2n}$$

Every factor equals $F(x)$

• Differentiate CDF to find the density of the maximum:

$$f_{X_{(n)}}(x) = \frac{d}{dx} \left(\frac{x^{2n}}{\theta^{2n}} \right) = \frac{2nx^{2n-1}}{\theta^{2n}}, \quad 0 < x \leq \theta$$

$$\bullet E(X_{(n)}) = \int_0^\theta x f_{X_{(n)}}(x) dx = \int_0^\theta x \left(\frac{2nx^{2n-1}}{\theta^{2n}} \right) dx = \frac{2n}{\theta^{2n}} \int_0^\theta x^{2n} dx = \frac{2n}{\theta^{2n}} \left(\frac{\theta^{2n+1}}{2n+1} \right)$$

$x \cdot x^{2n-1} = x^{2n}$

$$\int x^k dx = \frac{x^{k+1}}{k+1}$$

$$\int_0^\theta x^{2n} dx = \frac{\theta^{2n+1}}{2n+1}$$

$$E(X_{(n)}) = \frac{2n}{2n+1} \theta$$

$X_{(n)}$ Biased Estimator of θ

$$E(c\hat{\theta})=0 \rightarrow cE(\hat{\theta})=0 \rightarrow cE(X_{(n)}) \longrightarrow c\left(\frac{2n}{2n+1}\theta\right) = 0$$

$$\hat{\theta} = X_{(n)}$$

$$E(X_{(n)}) = \frac{2n}{2n+1} \theta$$

$$c\left(\frac{2n}{2n+1}\right) = 1$$

$$c = \frac{2n+1}{2n}$$

$$c\hat{\theta} = \frac{2n+1}{2n} X_{(n)}$$

Unbiased Estimator of θ

↓ continue

c) $F(\text{median}) = \frac{1}{2} = \frac{m^2}{\theta^2}$ plug in m for x
 $\underbrace{\quad}_m \times \theta^2$
 $\sqrt{\frac{\theta^2}{2}} = \sqrt{m^2}$

$$\frac{\theta}{\sqrt{2}} = m(\theta) \longrightarrow g(\theta) = \frac{\hat{\theta}}{\sqrt{2}} = \frac{X_{(n)}}{\sqrt{2}} = \hat{m}$$

$$\hat{\theta} = X_{(n)}$$

• Prove $\hat{m} \xrightarrow{P} m$, meaning for every $\epsilon > 0$, $P(|\hat{m} - m| > \epsilon) \rightarrow 0$ as $n \rightarrow \infty$

$$\hat{m} = \frac{X_{(n)}}{\sqrt{2}}, \quad m = \frac{\theta}{\sqrt{2}}$$

As sample size $\uparrow$, prob. that $\hat{m}$ differs substantially from the true median approaches 0.

$$P(|\hat{m} - m| > \epsilon) = P\left(\left|\frac{X_{(n)}}{\sqrt{2}} - \frac{\theta}{\sqrt{2}}\right| > \epsilon\right) = P\left(\left|\frac{X_{(n)} - \theta}{\sqrt{2}}\right| > \epsilon\right) = P(|X_{(n)} - \theta| > \epsilon\sqrt{2})$$

$$= P(\theta - X_{(n)} > \sqrt{2}\epsilon)$$

$$= P(X_{(n)} < \theta - \sqrt{2}\epsilon)$$

$$F(x) = P(X_i \leq x) = \frac{x^2}{\theta^2}$$

$$P(X_{(n)} < x) = [F(x)]^n$$

$$\text{Let } x = \theta - \sqrt{2}\epsilon$$

$$= [F(\theta - \sqrt{2}\epsilon)]^n = \left[\frac{(\theta - \sqrt{2}\epsilon)^2}{\theta^2}\right]^n = \left(1 - \frac{\sqrt{2}\epsilon}{\theta}\right)^{2n} \xrightarrow{\text{Approaches 0 as } n \uparrow} 0$$

$$0 < \left(1 - \frac{\sqrt{2}\epsilon}{\theta}\right)^{2n} < 1 \text{ for } 0 < \sqrt{2}\epsilon < \theta$$

$$\text{and } P(|\hat{m} - m| > \epsilon) \xrightarrow{\text{Approaches 0}} 0$$

$$\text{Hence } \hat{m} = \frac{X_{(n)}}{\sqrt{2}} \xrightarrow{P} \frac{\theta}{\sqrt{2}} = m$$

4. Suppose $X_1, X_2, \dots, X_n$ are iid with pdf $f(x; \theta) = \theta^{-1} e^{-x/\theta}$, $0 < x < \infty$, zero elsewhere. Find the mle of $P(X \leq 2)$ and show that it is consistent.

$$f(x; \theta) = \theta^{-1} e^{-x/\theta}$$

$$L(\theta) = \prod_{i=1}^n \frac{1}{\theta} e^{-x_i/\theta} = \theta^{-n} \exp\left(-\frac{1}{\theta} \sum_{i=1}^n x_i\right)$$

$$\ell(\theta) = -n \log \theta - \frac{1}{\theta} \sum_{i=1}^n x_i$$

$$\ell'(\theta) = -\frac{n}{\theta} + \frac{\sum_{i=1}^n x_i}{\theta^2} = 0$$

$$= -n\theta + \sum x_i = 0$$

$$n\theta = \sum x_i$$

$$\hat{\theta} = \frac{1}{n} \sum_{i=1}^n x_i = \bar{x}$$

$$\begin{aligned} P(X \leq 2) &= \int_0^2 \frac{1}{\theta} e^{-x/\theta} dx = 1 - e^{-2/\theta} \\ &= [-e^{-x/\theta}]_0^2 = e^{-2/\theta} - (-e^0) \\ &= 1 - e^{-2/\theta} = 1 - e^{-2/\bar{x}} \\ &\quad \hat{\theta} = \bar{x} \end{aligned}$$

$$P(X \leq 2) = 1 - e^{-2/\bar{x}}$$

Since $E(X_i) = \theta$

By the Law of Large Numbers: $\bar{x} \xrightarrow{P} E(X_i)$

Therefore, $\bar{x} \xrightarrow{P} \theta$

$g(t) = 1 - e^{-2/t}$ is continuous for $t > 0$, the Continuous Mapping Theorem:

$$1 - e^{-2/\bar{x}} \xrightarrow{P} 1 - e^{-2/\theta}$$

Thus, $1 - e^{-2/\bar{x}}$ is a consistent estimator for $P(X \leq 2)$